# Terrain-Sensitive Accessibility Modelling for Rural Settlements: A GIS–AHP Framework with Participatory Calibration

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**Abstract**

Rural accessibility in hilly regions presents unique challenges that conventional proximity-based measures fail to capture adequately. This study develops a comprehensive accessibility assessment framework that integrates Geographic Information Systems (GIS) with the Analytic Hierarchy Process (AHP) to address the complex interplay of terrain constraints and service priorities in mountainous rural areas. The methodology was applied to the West Garo Hills district in Meghalaya, India, encompassing 1,176 habitations serving 501,294 residents across challenging topographic conditions. A structured stakeholder survey across 49 respondents derived consistency-validated priority weights for both service types and travel factors. Spatial analysis utilised high-resolution Digital Elevation Models to generate slope-adjusted friction surfaces, integrated with road network data to compute realistic travel-time rasters. Machine learning-enhanced analysis of Sentinel-2 satellite imagery achieved 89.7% accuracy in built-up area identification for habitation boundary extraction. Results reveal extreme spatial disparities in accessibility, with scores ranging from 0.130 to 0.642 across habitations—a 4.95-fold difference indicating severe inequity in service access. Notably, 57.7% of habitations fall into low or very low accessibility categories, while only 2.3% achieve very high accessibility scores. The analysis demonstrates an inverse relationship between settlement size and accessibility, with larger habitations experiencing lower access despite housing greater populations. This finding suggests that population concentration has not translated into proportional infrastructure investment, creating compound disadvantages for 272,260 residents living in low-accessibility areas. The AHP-GIS framework provides a participatory, terrain-aware approach to accessibility assessment that captures both physical constraints and community priorities, offering actionable insights for infrastructure planning and resource allocation in hilly rural regions.

**Keywords**

Accessibility index, AHP, GIS, Rural development, Hilly terrain, Infrastructure planning, Participatory planning, Terrain-sensitive modelling

# 1. Introduction

Providing equitable, reliable access to essential infrastructure—healthcare facilities, schools, public transport, and markets—is foundational to inclusive rural development. Such access shapes socioeconomic outcomes by enabling education, improving health service use, and facilitating market integration (Sapkota, 2018; Briceño-Garmendia et al., 2015). In India's hilly, topographically complex regions—especially the Northeast—provision is constrained by geography and institutional challenges. Steep slopes, forest cover, scattered settlements, and underdeveloped road networks create severe accessibility deficits (Tobler, 1993; Shrestha et al., 2014; Kapoor & Bansal, 2021). While national programs like the Pradhan Mantri Gram Sadak Yojana (PMGSY) have improved road penetration, last-mile connectivity and terrain-responsive planning remain critical gaps (World Bank, 2016).

Rural settlements often grow organically without formal land-use controls, especially in tribal or remote administrative blocks. Infrastructure provisioning lags behind habitation expansion, leaving populations underserved despite geographic proximity to services (Bosch & Ferguson, 2007). This mismatch exposes the limitations of conventional accessibility metrics, which typically measure linear distance to roads or services but fail to account for travel friction from slope, road quality, surface type, and seasonal disruptions (Geurs & van Wee, 2004; Shrestha, 2018).

## *1.1. Background and Motivation*

Equitable access to healthcare, education, markets, and government facilities is critical for socio-economic inclusion and quality of life in rural, hilly regions (Briceño-Garmendia et al., 2015; Sapkota, 2018). Inaccessibility deepens poverty, limits welfare access, and restricts economic participation, especially among women, the elderly, and marginalised groups (World Bank, 2023). Despite major infrastructure initiatives like PMGSY extending rural road networks, actual travel remains shaped by challenging terrain, inconsistent road quality, and seasonal disruptions—especially in Meghalaya (Kapoor & Bansal, 2021; Nautiyal & Sharma, 2021).

Traditional metrics like the Rural Accessibility Index (RAI) offer useful overviews but rely on binary thresholds (e.g., presence of an all-weather road within 2 km) that ignore real terrain challenges (World Bank, 2006). Steep gradients, landslide-prone corridors, and poor surfacing mean 2 km can require over an hour of travel (Shrestha et al., 2014). Proximity-based models thus overestimate true accessibility (Geurs & van Wee, 2004; Shrestha, 2018).

Furthermore, infrastructure planning has largely been top-down, shaped by administrative priorities or expert judgment that often overlooks lived experience. Local perspectives—like which services are most critical, or how slope or safety affects travel—are rarely quantified or incorporated (De Marinis & Sali, 2020). This misalignment leads to under-targeted investments and ineffective interventions.

To address these gaps, this study proposes an integrated approach combining local user perspectives with advanced spatial modelling. Using Analytic Hierarchy Process (AHP) surveys calibrated by residents and GIS-based cost-distance modelling, it computes a composite accessibility index reflecting both subjective priorities and objective terrain constraints. This dual-tool methodology supports habitation-level planning and provides a replicable model for terrain-sensitive assessment in hilly rural regions.

### *1.1.1. Land Use and Mobility Linkages*

Land use and accessibility in rural hilly layouts are deeply intertwined. Failure to account for slope, terrain constraints, and institutional barriers results in fragmented infrastructure that doesn't address real mobility challenges (Anderson et al., 1976; Jedlička et al., 2019). Proximity to transport infrastructure shapes development patterns, housing, and service provision. Handy (2002) distinguished between mobility and accessibility in transport planning, underscoring the need for spatial equity. Rimal et al. (2018) demonstrated that LU/LC change in urban and peri-urban areas is heavily influenced by road network growth and connectivity.

## *1.2. Deficiencies in Existing Models and Policy Gaps*

Recent GIS-based approaches have improved rural accessibility measurement (Katpatal & Thorat, 2022; Malczewski, 1999) but often use uniform weights at regional scales, ignoring local priorities and realities. Widely used models like the RAI focus on physical proximity to all-weather roads but overlook experiential factors like slope-induced walking effort or road condition-related delays (World Bank, 2016). This is especially problematic in hilly areas where even short distances can mean long travel times.

Lack of comprehensive, participatory planning frameworks compounds disparities. Poor agency coordination, delayed implementation, and underutilisation of spatial decision support tools inhibit rural development program effectiveness (Sengupta & Banerji, 2012; Nallathiga, 2009).

## *1.3. Study Contribution and Organisation*

This study addresses these limitations by proposing a user-calibrated, habitation-level accessibility index tailored for hilly regions. The methodology combines stakeholder-informed weights using the Analytic Hierarchy Process (AHP) (Saaty, 2008) with high-resolution GIS spatial analysis to produce a topographically aware, locally contextualised index. An AHP survey in West Garo Hills, Meghalaya, involved 49 participants rating the relative importance of multiple criteria (e.g., distance, slope, travel time, safety) and key services (e.g., hospitals, schools, markets). These subjective weights were applied to GIS-generated travel surfaces incorporating terrain slope, road type, and impedance factors. The resulting continuous, normalised accessibility scores enable planners to identify underserved areas and prioritise infrastructure investments.

This paper is structured as follows: a review of relevant literature on accessibility modelling and GIS-AHP integration; description of the study area and data sources; detailed methodology including survey design, weight computation, GIS processing, and index derivation; presentation of results with thematic maps and rankings; discussion of policy and planning implications; and finally, conclusions and recommendations.

### *1.3.1. Analytic Hierarchy Process (AHP) for Participatory Weighting*

The Analytic Hierarchy Process (AHP) provides a structured framework for integrating user preferences into accessibility modelling. Developed by Saaty (1980), AHP is widely used in infrastructure planning where multiple qualitative and quantitative criteria must be prioritised (Ishizaka & Labib, 2011). In this study, it supports assigning relative importance to factors like distance, travel time, slope, road quality, safety, and service type. Its pairwise comparison method offers a consistent, transparent basis for incorporating human judgment (Vaidya & Kumar, 2006).

Stakeholder surveys used 9-point pairwise matrices. Responses were aggregated using geometric means, normalised into weight vectors, and validated using Consistency Ratio (CR) metrics (Saaty, 2008). This participatory approach ensures resulting weights reflect local priorities, offering a grounded alternative to expert-only or literature-derived values. The final AHP weights guide prioritisation of service accessibility and travel impedance criteria in constructing the composite index.

### *1.3.2. Geographic Information Systems (GIS) for Terrain-Aware Accessibility Modelling*

GIS is critical for modelling terrain-aware accessibility in remote, rugged regions (Goodchild, 2000; Longley et al., 2015). High-resolution Digital Elevation Models (DEMs), such as SRTM, were used to derive slope rasters (USGS, 2015; Grohmann, 2004). Road network data came from OpenStreetMap (OpenStreetMap contributors, 2025) and field surveys, classified by surface condition to capture realistic travel impedance (Jiang & Claramunt, 2004).

GIS-based cost-distance analysis computed travel-time rasters from each habitation to key services. These rasters combined slope-derived friction surfaces with road segment speeds and resistance values (Wang et al., 2008). Service locations—including schools, clinics, markets, and transport nodes—were geocoded as destinations. Travel surfaces were standardised using min-max scaling and combined with AHP-derived weights through raster algebra (Malczewski, 1999), producing a normalised accessibility index. This enables nuanced identification of low-access habitations for data-driven planning.

**Objectives**

The overall aim of this study was to develop a comprehensive GIS-AHP-based accessibility assessment framework that integrates terrain-sensitive spatial modelling with community-prioritised service weights to address the complex challenges of infrastructure planning in hilly rural regions.

To achieve this aim, four key challenges had to be overcome. These challenges are closely related to the four main objectives of this study:

i. To develop a participatory methodology for deriving locally-relevant accessibility weights using the Analytic Hierarchy Process (AHP) that captures community priorities and service preferences through structured stakeholder engagement.
ii. To create terrain-aware travel impedance models that account for slope-induced friction and realistic travel conditions in mountainous environments, moving beyond traditional proximity-based measures.
iii. To implement machine learning-enhanced habitation mapping for accurate spatial analysis and automated identification of inhabited areas using satellite imagery and spectral analysis techniques.
iv. To demonstrate the framework's practical application through a comprehensive accessibility assessment of West Garo Hills district, Meghalaya, providing empirical validation and actionable insights for rural infrastructure planning and resource allocation.

## 2. Literature Review

### *2.1. Understanding Rural Accessibility in Complex Terrain*

Accessibility in rural contexts is broadly defined as the ease with which residents can reach essential services such as healthcare, education, markets, and administrative centres. However, in mountainous and topographically rugged regions, physical accessibility is not simply a function of distance but a complex interplay of terrain, road quality, and travel effort (Geurs & van Wee, 2004; Páez et al., 2012). Conventional metrics, such as the Rural Accessibility Index (Roberts et al., 2006), emphasise proximity to all-weather roads but fail to incorporate ground realities in steep and ecologically sensitive landscapes where 2 km of walking may translate into over an hour of travel due to slope and poor paths. These terrain-induced accessibility challenges are well-documented in hilly regions globally, particularly where steep gradients and poor infrastructure intersect to create compound mobility barriers (Shrestha et al., 2014; Kapoor & Bansal, 2021), as illustrated in Fig. 1.

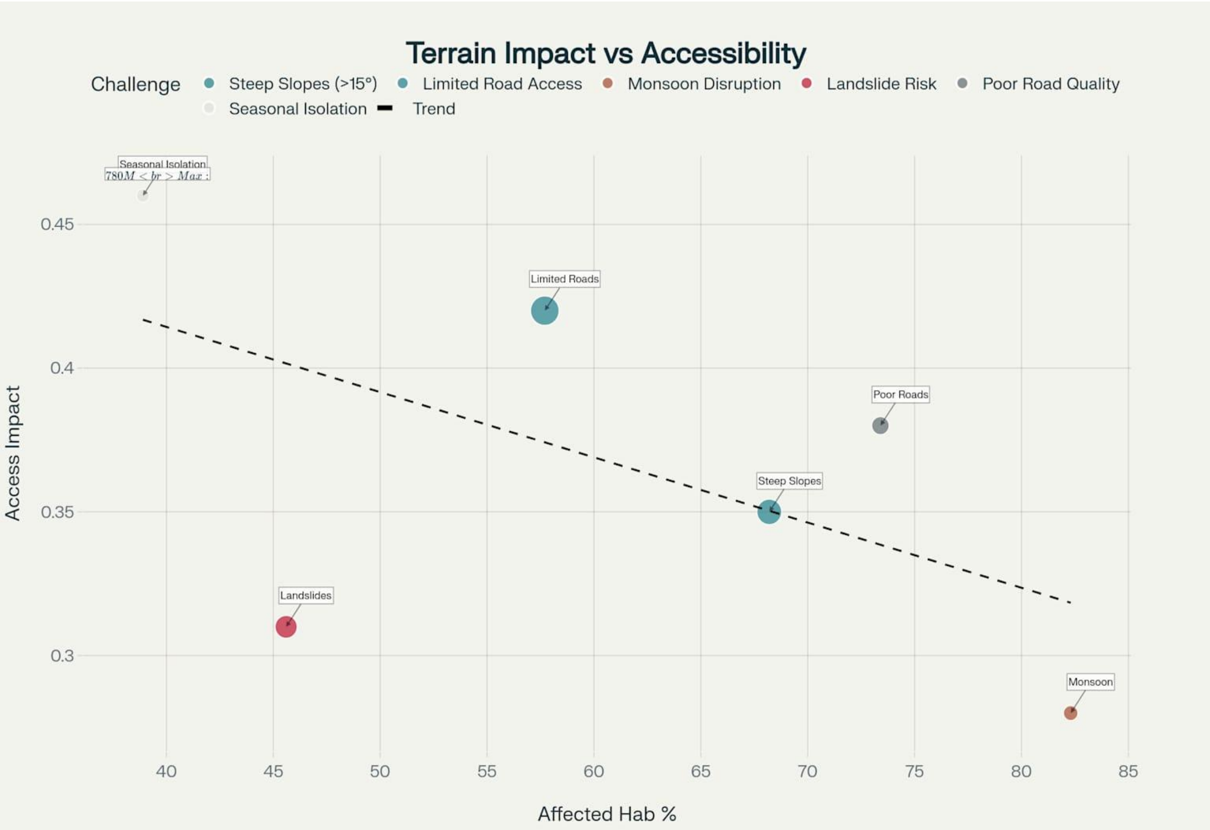


**Fig. 1**. Terrain Challenges Impact on Rural Accessibility in Hilly Regions

Furthermore, spatial exclusion persists in regions where infrastructure planning does not reflect the lived experience of rural dwellers. In regions like Northeast India, natural barriers such as steep gradients, forest cover, and monsoon-related disruptions make road-based service access highly variable (Singh et al., 2013). These conditions require a move from proximity-based indicators to dynamic, participatory models that reflect both travel difficulty and service prioritisation.

### *2.2. GIS in Terrain-Sensitive Accessibility Assessment*

Geographic Information Systems (GIS) are increasingly used to simulate terrain-sensitive accessibility through raster-based cost-distance analysis and travel impedance modelling (Longley et al., 2015; Bell & Wilson, 2006). Digital Elevation Models (DEMs), such as those from the SRTM or ASTER datasets, are commonly used to derive slope rasters, which are then integrated with road and land-use layers to model travel time more accurately (Grohmann, 2004; Sasidharan et al., 2020).

Recent studies have modelled walking time to health facilities and schools using slope-adjusted friction surfaces in Nepal (Shrestha, 2018), Kenya (Noor et al., 2006), and Ethiopia (Tsegaye et al., 2020), demonstrating how GIS can quantify the spatial burden of terrain on access. In India, GIS-based approaches have been adopted in Sikkim and Uttarakhand for identifying "access-poor" villages and optimising rural road networks (Bisht & Mishra, 2021; Nautiyal & Sharma, 2021).

Such methods provide spatial granularity, enabling decision-makers to assess disparities at the habitation level rather than at block or district scales, which often mask localised deprivation (Ebener et al., 2005).

### *2.3. Analytic Hierarchy Process (AHP) for Weight Calibration*

The Analytic Hierarchy Process (AHP), introduced by Saaty (1980), is a multi-criteria decision-making (MCDM) method used extensively in infrastructure planning to prioritise alternatives under complex trade-offs. AHP uses pairwise comparisons and consistency checks to quantify subjective preferences of stakeholders into usable numerical weights (Ishizaka & Labib, 2011). In the context of rural accessibility, it allows residents and planners to rank the importance of factors such as road condition, travel time, slope, cost, and safety—elements that are difficult to capture in conventional models.

Studies by Pal and Ghosh (2016) in West Bengal and Rao and Srinivas (2020) in Karnataka applied AHP in road alignment and infrastructure prioritization by involving community preferences. De Marinis and Sali (2020) applied AHP-GIS models for transport accessibility assessment in Italy, underscoring how participatory methods enhance local relevance and planning outcomes. Similarly, Jat et al. (2017) demonstrated the integration of AHP with spatial layers to map urban accessibility in India, but few studies apply this to rural, hilly settings at fine resolution.

### *2.4. Combined AHP-GIS Frameworks: A Participatory Spatial Model*

The integration of AHP with GIS creates powerful participatory spatial models that address both user priorities and terrain complexities. Malczewski (1999) outlined early frameworks for multi-criteria spatial decision support systems, which have since evolved into highly adaptable models for rural accessibility planning. Studies such as Chen et al. (2019) used AHP-GIS models to assess service accessibility in mountainous China, while Mugo and Odera (2019) applied similar techniques in rural Kenya.

In India, Katpatal and Thorat (2022) proposed a Road Infrastructure Development Index using AHP-GIS integration for urban layouts, arguing for its utility in data-sparse environments. Although their work focused on urban design, the core methodology is transferable to rural and hilly regions. The major gap remains: most AHP-GIS models do not compute habitation-level indices or use actual stakeholder surveys to calibrate weights.

The study fills this gap by using survey-driven AHP weight derivation from 1,176 habitations and coupling it with GIS-based travel time surfaces to construct a composite, normalised accessibility index. The use of sigmoid-based slope normalization (Wang et al., 2020) further refines the terrain penalty modeling and enhances realism in friction surface calculations.

### *2.5. Policy, Planning, and Equity Considerations*

The need for decentralised and user-responsive infrastructure planning is emphasised by recent planning literature (Sengupta & Banerji, 2012; Chandira, 2007). The UN SDGs, particularly Goal 9.1 (develop quality, reliable, sustainable infrastructure) and Goal 11.2 (provide access to safe, affordable transport for all), demand localised and data-driven approaches that reflect lived inequities.

Empirical evidence shows that lack of accessibility correlates with higher maternal mortality, lower school attendance, and poor economic resilience in rural India (Dasgupta et al., 2022; Sapkota, 2018). Tools like the one proposed in this study, participatory, spatially refined, and terrain-aware, are necessary to complement top-down schemes such as PMGSY and make last-mile connectivity a planning reality rather than a statistical abstraction.

Participatory planning approaches recognise that successful rural infrastructure development requires active community involvement throughout the planning, implementation, and maintenance processes (Cornwall & Jewkes, 1995; Pretty, 1995). Recent studies highlight that infrastructure projects in rural areas achieve better outcomes when they incorporate local knowledge and priorities through structured participation frameworks (Arnstein, 1969; Cornwall & Jewkes, 1995). The involvement of community members ensures that development initiatives address actual needs rather than perceived needs as determined by external planners (Pretty, 1995).

**Summary of Gaps in Literature Addressed by This Study**

| **Aspect** | **Existing Literature** | **Gap** | **This Study's Contribution** |
|---|---|---|---|
| Accessibility Metric | Road proximity (RAI), Euclidean distance | No travel impedance or slope modeling | GIS-based cost-distance with slope and surface quality |
| User Involvement | Expert judgment or policy-driven weights | Lack of participatory calibration | AHP survey with 49 habitations |
| Spatial Resolution | Regional or block scale | Ignores habitation-level disparities | Index computed for individual habitations |
| Terrain Sensitivity | Limited to DEM-derived slope | No normalization or real-world travel impact | Sigmoid-normalized slope with z-scaling |
| Service Diversity | Mostly school/healthcare focused | Lacks integrated multi-service accessibility | Models access to healthcare, education, transport, and markets |
| Validation | Rarely conducted | No consistency checks or stakeholder grounding | CR analysis, ranking maps, and spatial validation outputs |

## 3. Materials and Methods

### *3.1. Study Area*

#### *3.1.1. Regional Overview*

West Garo Hills is one of the eleven districts in the state of Meghalaya, located in the western part of the Garo Hills region as shown in Fig, 2. Tura is the district headquarters and also the region's primary urban and administrative centre. The district spans an area of approximately 3,677 sq. km and is characterised by rugged topography, interspersed with valleys, rivers, and dense forest cover.

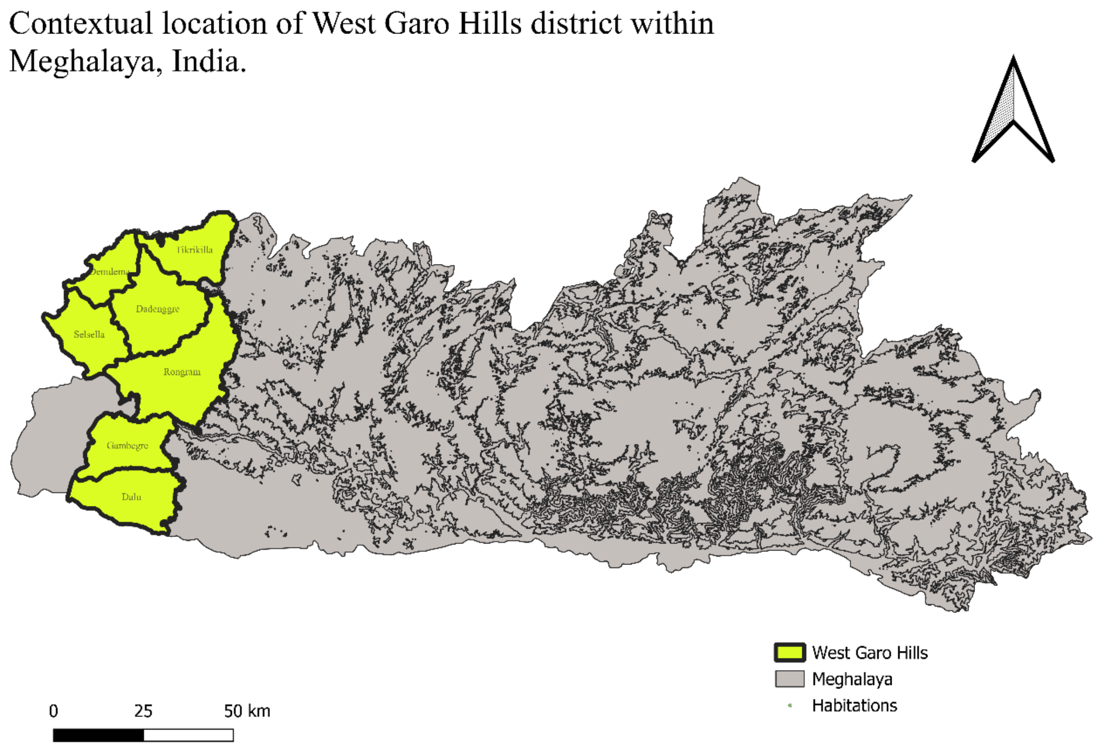


**Fig. 2**. Map of the contextual location of West Garo Hills

*3.1.2. Terrain and Climatic Challenges*

The district's hilly terrain significantly impacts transport and accessibility. Elevation varies widely, and steep gradients are common, making travel difficult, especially during the monsoon season when roads often become impassable. Seasonal landslides and waterlogging further aggravate mobility issues. These geographic constraints have historically hindered infrastructure development and service delivery.

*3.1.3. Demographic And Settlement Characteristics*

West Garo Hills has a predominantly rural population, with scattered habitation patterns and limited nodal connectivity. The majority of the residents belong to the Garo tribe and engage in agriculture or allied activities. Settlements range from small villages to larger habitation clusters such as Dadenggre, Rongram, and Tikrikilla. These areas have varied access levels to services like schools, health centres, markets, and transport nodes.

*3.1.4. Need for Accessibility Assessment*

Despite being relatively close to road networks or service centres in terms of straight-line distance, many habitations experience poor real-world accessibility due to factors such as slope, road quality, and lack of public transport. Conventional planning models often overlook these micro-level challenges, making habitat-scale assessment essential. The district thus provides a suitable case for applying an AHP-GIS-based accessibility model that integrates both user perception and spatial friction.

*3.1.5. GIS Data Coverage*

For this study, spatial layers such as administrative boundaries, road networks, Digital Elevation Model (DEM), and service facility locations were compiled and processed in QGIS (QGIS Development Team, 2025). High-resolution slope rasters were generated and used as key frictional inputs to compute travel-time surfaces. All analyses were carried out using the coordinate reference system EPSG:32646 (WGS 84 / UTM Zone 46N) as per the EPSG Geodetic Parameter Dataset (2024), ensuring spatial accuracy.

*3.2. Data Collection*

High-resolution spatial datasets were compiled from multiple authoritative sources to support comprehensive accessibility analysis. Table 1 provides a detailed summary of the datasets used in this research, their specifications, sources, and applications in the methodology.

**Table 1: Data Collection Summary**

| **Dataset** | **Spatial/Temporal Resolution** | **Format/Geometry** | **Source** | **Data Use** |
|---|---|---|---|---|
| SRTM DEM v3.0 | 30 m; 2000 epoch | GeoTIFF/Raster | NASA/USGS EROS | Terrain slope analysis and travel impedance modeling |
| Sentinel-2 L2A | 10-60 m; Nov 2024-May 2025 | GeoTIFF/Raster | ESA Copernicus Hub | Machine learning-based habitation boundary extraction |
| Administrative Boundaries | Village/Block polygons | Shapefile/Polygons | PMGSY GIS Portal | Spatial analysis units and population attribution |

| Dataset | Spatial/Temporal Resolution | Format/Geometry | Source | Data Use |
|---|---|---|---|---|
| Road Network | Variable, 2025 extract | Shapefile/Lines | OpenStreetMap | Travel network modelling and surface classification |
| Service Locations | Point coordinates | Shapefile/Points | Field surveys + Government databases | Accessibility destination mapping |
| Population Data | Village-level; Census 2011 | Tabular/CSV | Census of India | Population-weighted accessibility analysis |
| Stakeholder Survey Data | 49 respondents; Jan-Feb 2025 | Tabular/Excel | Primary field campaign | AHP pairwise comparison matrices |

The Shuttle Radar Topography Mission (SRTM) Digital Elevation Model with 30-meter resolution was obtained from the USGS Earth Resources Observation and Science Centre. Administrative boundary shapefiles for villages, blocks, and district extents were acquired from the Pradhan Mantri Gram Sadak Yojana (PMGSY) database (Ministry of Rural Development, 2025), ensuring compatibility with official administrative units. Road network data was extracted from OpenStreetMap and systematically validated through field surveys and GPS tracking campaigns conducted between January and March 2025.

Service facility locations, including healthcare centres, educational institutions, markets, administrative offices, and transportation nodes, were geocoded through a combination of government databases, field surveys, and satellite imagery analysis. Cloud-free Sentinel-2 Level-2A satellite imagery covering the study period (Drusch et al., 2012) from November 2024 to May 2025 was acquired from the ESA Copernicus Hub. All spatial datasets were reprojected to UTM Zone 46N (EPSG:32646) coordinate reference system to ensure geometric accuracy (EPSG Geodetic Parameter Dataset 2024) during spatial analysis operations.

### *3.3. Analytical Techniques*

The GIS-based analytical framework developed in this study integrates stakeholder-calibrated Analytic Hierarchy Process (AHP) weights with terrain-sensitive spatial modelling to produce comprehensive accessibility indices at the habitation level. Fig. 3 presents the complete analytical framework adopted in this study, illustrating the integration of participatory weighting with spatial analysis. The methodology addresses key limitations of traditional proximity-based approaches by incorporating both community priorities and realistic travel impedance modelling in challenging topographic environments.

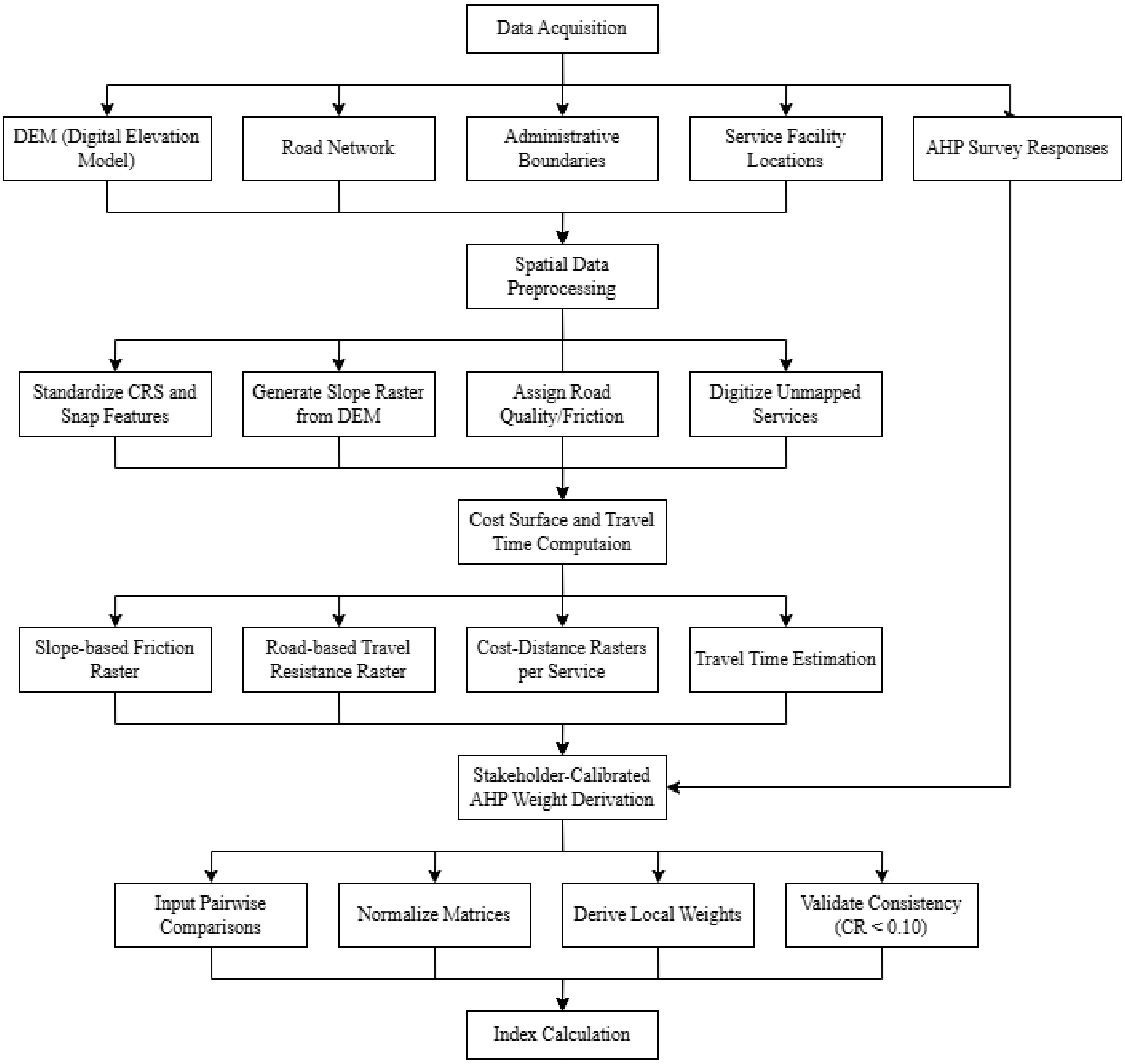


**Fig. 3**. Complete flowchart of the adopted methodology

### *3.3.1. Terrain-Sensitive Travel Impedance Modelling*

Slope-based travel impedance was modelled using a non-linear penalty function that reflects empirical relationships between terrain gradient and human locomotion effort in mountainous environments. Rather than treating slope as a separate accessibility criterion, it was incorporated as a friction multiplier in travel-time calculations to avoid methodological double-counting of terrain effects.

The district's varied topography, with elevations ranging from valley floors to hill peaks, significantly impacts travel patterns and accessibility. The terrain characteristics of the study area are visualised in Fig. 4, which illustrates the slope variations and topographic complexity that influence mobility patterns.

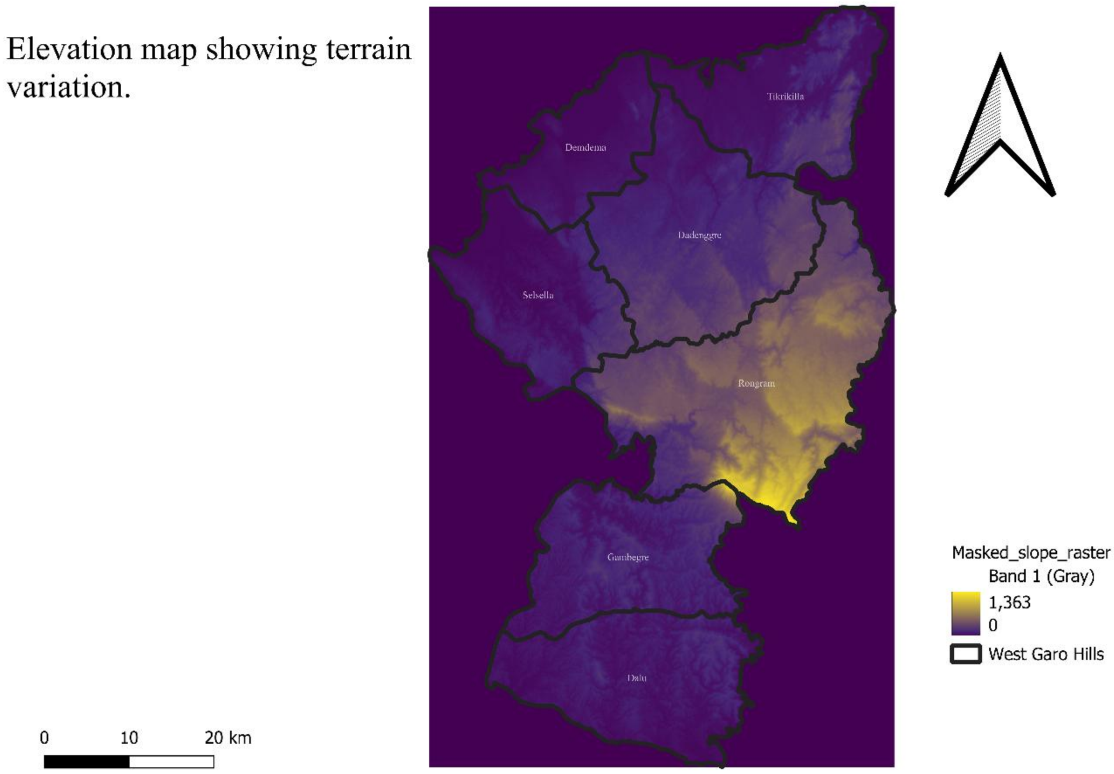


**Fig. 4**. Hillshade map visualising slope and terrain of WGH

To penalise steeper slopes, a sigmoid function was applied to the slope raster (Wang et al. 2008), defined as:

$$f(s) = \frac{1}{1 + e^{[-\alpha(s-s_0)]}}$$

where s represents local slope in degrees, α = 0.002 controls the steepness of the penalty curve, and $s_0$ = 15° serves as the inflexion point. This formulation assigns minimal impedance to gentle slopes (< 10°) while exponentially increasing travel resistance beyond 15 degrees.

Travel-time surfaces were generated using the GRASS GIS r.walk module (GRASS Development Team, 2025), which implements anisotropic cost-distance algorithms accounting for both slope and surface friction. Road segments were classified by surface type and assigned travel speeds based on field observations: paved roads (40 km/h), gravel roads (25 km/h), and earth tracks (10 km/h). Off-road areas received friction values based on land cover type and slope conditions.

### *3.3.2. Habitation Extraction from Satellite Imagery*

Accurate identification of built-up zones was a critical step in ensuring that the accessibility index reflected actual inhabited areas. Instead of relying on administrative village boundaries or buffers, this study employed a hybrid remote sensing and machine learning approach to extract habitation zones using Sentinel-2 Level-2A imagery.

The workflow involved selecting cloud-free Sentinel-2 surface reflectance imagery (November 2024–May 2025) composited with a median reducer to represent the dry season. Spectral indices NDVI (Rouse et al., 1974), NDBI, and Urban Index were calculated from relevant bands (B2, B3, B4, B8, B11, B12). In Google Earth Engine, a Random Forest classifier (50 trees) was trained on labelled samples (built-up, vegetation, water, barren) to produce a land-cover map, from which a binary habitation mask isolating built-up pixels was derived. Post-processing in QGIS used GDAL's Sieve algorithm to remove speckles (<8 pixels) and polygonise the mask into vector format. Proximity filtering retained only polygons within 3 km of surveyed habitation points, after which nearest-neighbour spatial joins assigned each habitation polygon its corresponding accessibility score. This machine learning approach resulted in high-resolution, terrain-aligned habitation boundaries that provide a realistic spatial foundation for accessibility assessment. Fig. 5 demonstrates the habitation boundaries extracted through Google Earth Engine analysis, showing the distribution of inhabited areas across the district's challenging terrain.

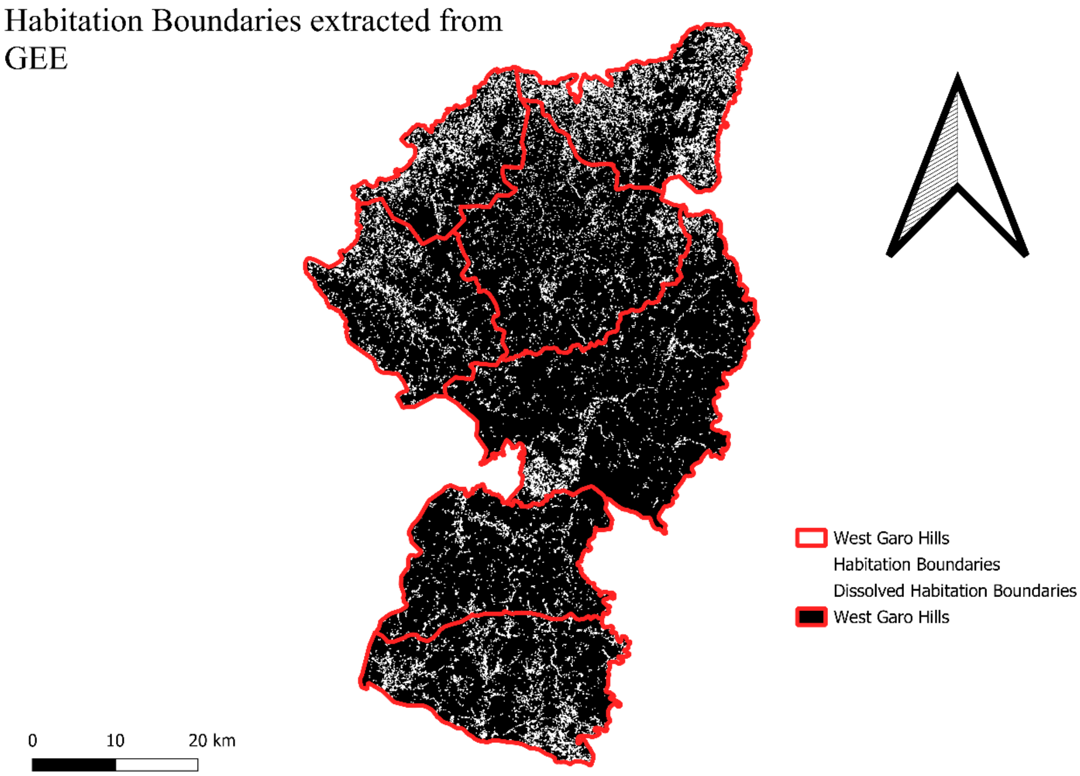


**Fig. 5**. Map showing Habitation boundaries extracted from GEE

This method resulted in high-resolution, terrain-aligned habitation boundaries, offering a realistic spatial foundation for the accessibility model and ensuring that accessibility scores are spatially attributed only to actual inhabited clusters.

*3.3.3. Stakeholder-Calibrated AHP Weighting Methodology*

A structured stakeholder engagement process was conducted to derive locally-relevant weights for accessibility assessment using the Analytic Hierarchy Process. The survey design employed Saaty's pairwise comparison methodology, where participants evaluated the relative importance of service types and travel factors using the standard 9-point scale.

Forty-nine respondents were selected through stratified sampling across administrative blocks, ensuring geographic representation and demographic diversity. The sample included village headmen, self-help group leaders, government officials, and community members representing different age groups and mobility needs. The survey instrument incorporated two hierarchical comparison matrices:

1. Service Category Priorities: Healthcare facilities, educational institutions, markets, administrative offices, and transportation nodes

2. Travel Factor Importance: Distance, travel time, slope difficulty, road surface condition, and safety considerations

Individual pairwise comparison matrices were validated using Saaty's Consistency Ratio (CR), with matrices achieving CR < 0.10 accepted for analysis. Geometric mean aggregation was employed to generate group consensus weights, preserving the reciprocal property of pairwise comparisons. The principal eigenvector method was used to derive normalised priority weights from aggregated matrices:

$$w_j = \frac{\lambda_{max,j}}{\sum_{j=1}^{n} \lambda_{max,j}}$$

where $\lambda_{max,j}$ represents the principal eigenvalue for criterion j

*3.3.4. Cost-Distance Travel Time Analysis*

Comprehensive travel-time modelling was implemented using multi-criteria cost-distance analysis that integrates terrain impedance with infrastructure constraints. Cost surfaces were constructed by combining slope-adjusted friction values with road network impedance. The cost-distance analysis computed cumulative travel time rasters for each service category:

$$T_{ij} = \sum_{k=1}^{n} \frac{d_k}{v_k} \times P(s_k)$$

where $T_{ij}$ represents travel time from habitation i to service j, $d_k$ is the distance of path segment k, $v_k$ is the travel speed for segment k, and $P(s_k)$ is the slope penalty for segment k. Road network quality and surface conditions represent critical factors in determining realistic travel times, particularly in hilly terrain where surface degradation is common. Fig. 6 illustrates the road network classification and surface quality distribution across West Garo Hills, highlighting the variation in infrastructure conditions that affect accessibility.

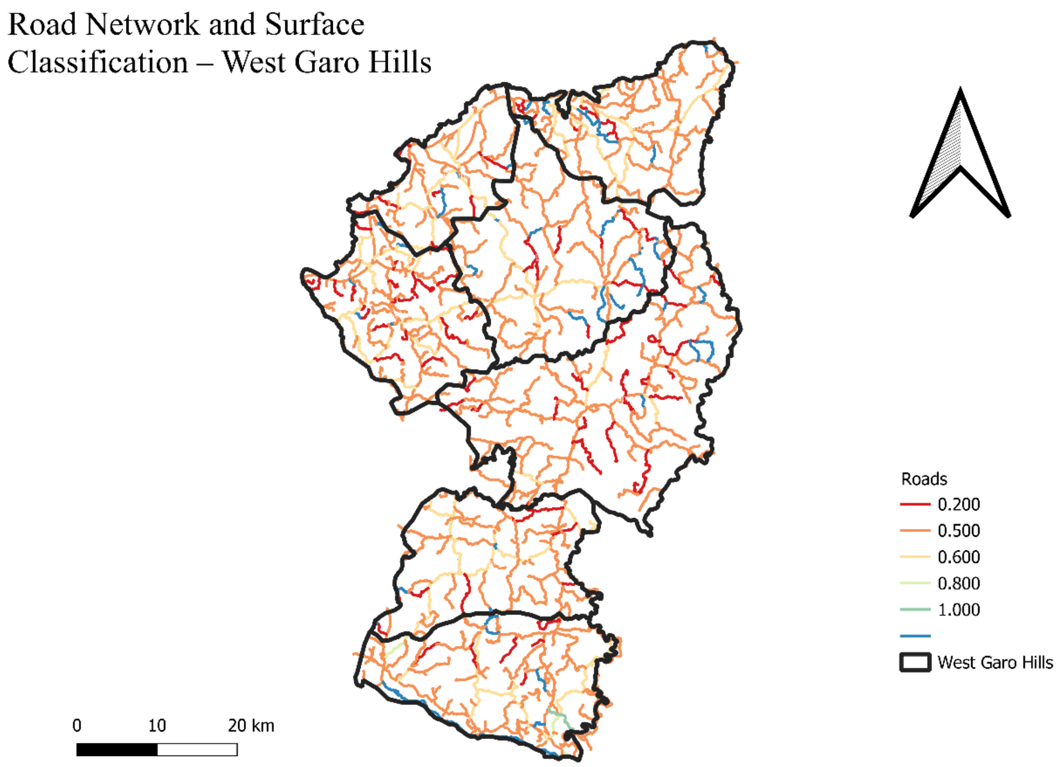


**Fig. 6**. Map visualising Road Networks and Road Surface Quality of WGH

*3.3.5. Composite Accessibility Index Derivation*

The composite accessibility index was computed through a weighted linear combination of normalised travel-time surfaces. Each travel-time surface underwent min-max normalisation:

$$X_{norm,ij} = \frac{T_{max,j} - T_{ij}}{T_{max,j} - T_{min,j}}$$

The composite accessibility index was calculated using AHP-derived weights:

$$AI_i = \sum_{j=1}^{5} w_j \times X_{norm,ij}$$

where $AI_i$ represents the accessibility index for habitation i, $w_j$ denotes the AHP weight for service category j, and $X_{norm,ij}$ represents the normalised accessibility score.

## 4. Results

*4.1. Overall Accessibility Distribution*

The composite Accessibility Index (AI) for the 1,114 habitations in West Garo Hills spans a range of 0.130 – 0.642, indicating a 4.95-fold disparity between the least and most accessible communities. Spatial patterns of accessibility across West Garo Hills reveal distinct zonal variations in service access, with clear concentration effects around administrative centres. Fig. 7 presents a Voronoi plot visualization of accessibility zones throughout the district, demonstrating the spatial heterogeneity in access conditions.

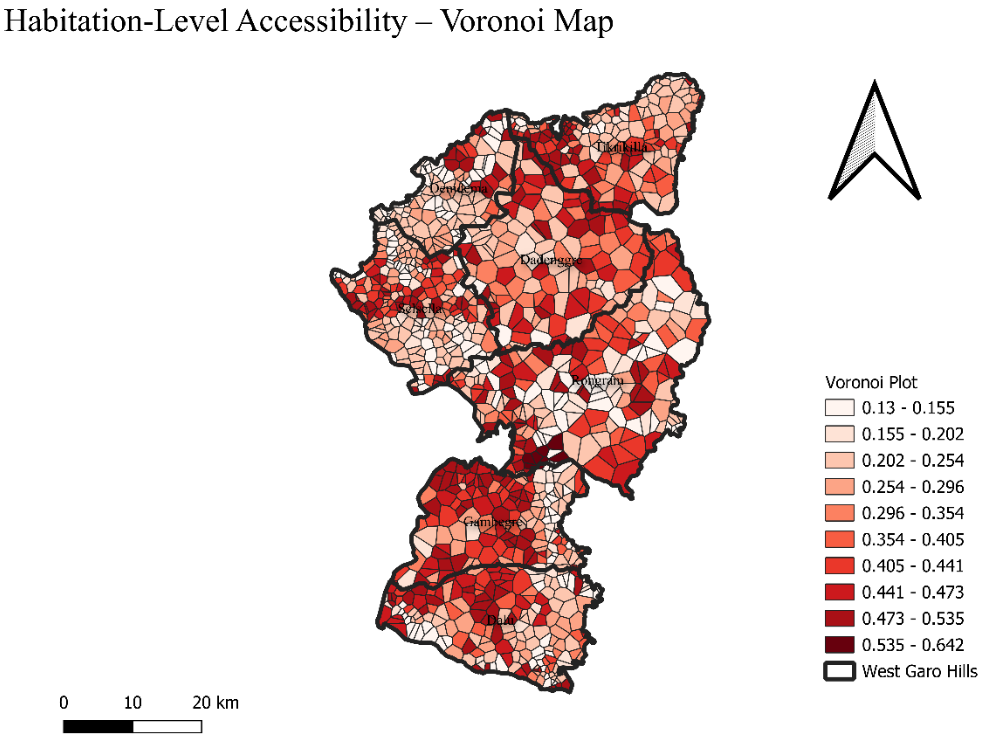


**Fig. 7**. Voronoi Plot representing Zonal Accessibility throughout WGH

The accessibility analysis reveals significant spatial clustering of both high and low accessibility areas, forming distinct hotspots and cold spots across the district. Fig. 8 provides a heatmap representation of accessibility variations and identifies accessibility hotspots, showing the pronounced spatial inequalities in service access.

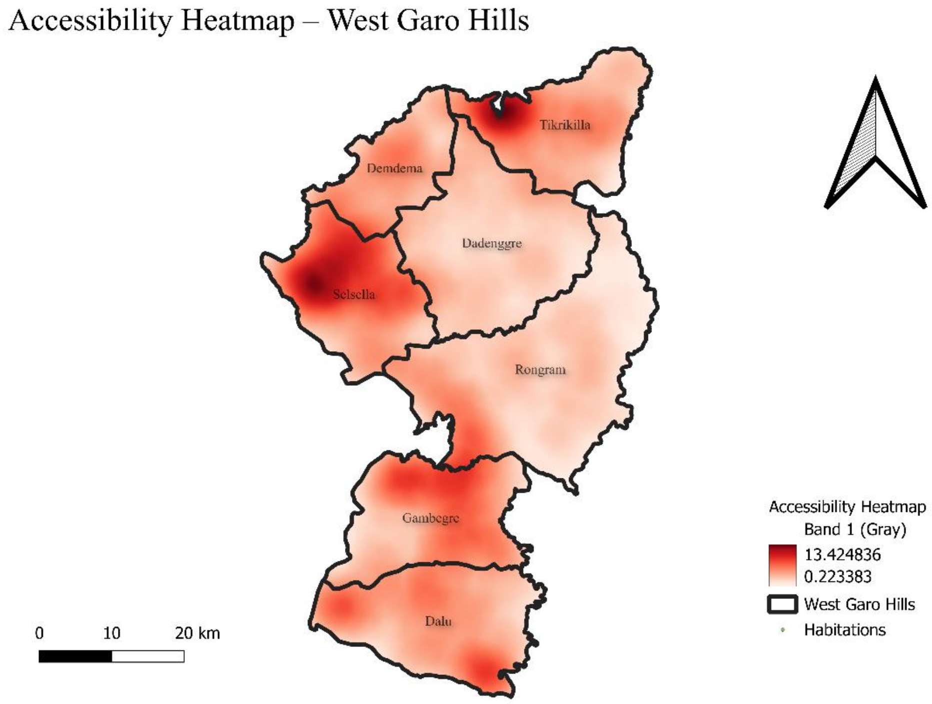


**Fig. 8**. Heatmap representing change in accessibility and Hotspots in WGH

The mean AI is 0.305 with a standard deviation of 0.128, producing a coefficient of variation of 42.1%, which confirms substantial heterogeneity in service access across District. The distribution is positively skewed: 75% of habitations record scores below 0.437, while only 2.3% exceed 0.535, highlighting a pronounced long-tail of well-served settlements centred on Tura and its peri-urban satellites. The distribution of habitations across accessibility categories reveals a highly skewed pattern, with the majority of settlements experiencing below-average access conditions. Fig. 9 illustrates the distribution of habitations by accessibility level, demonstrating the prevalence of low accessibility conditions across the district.

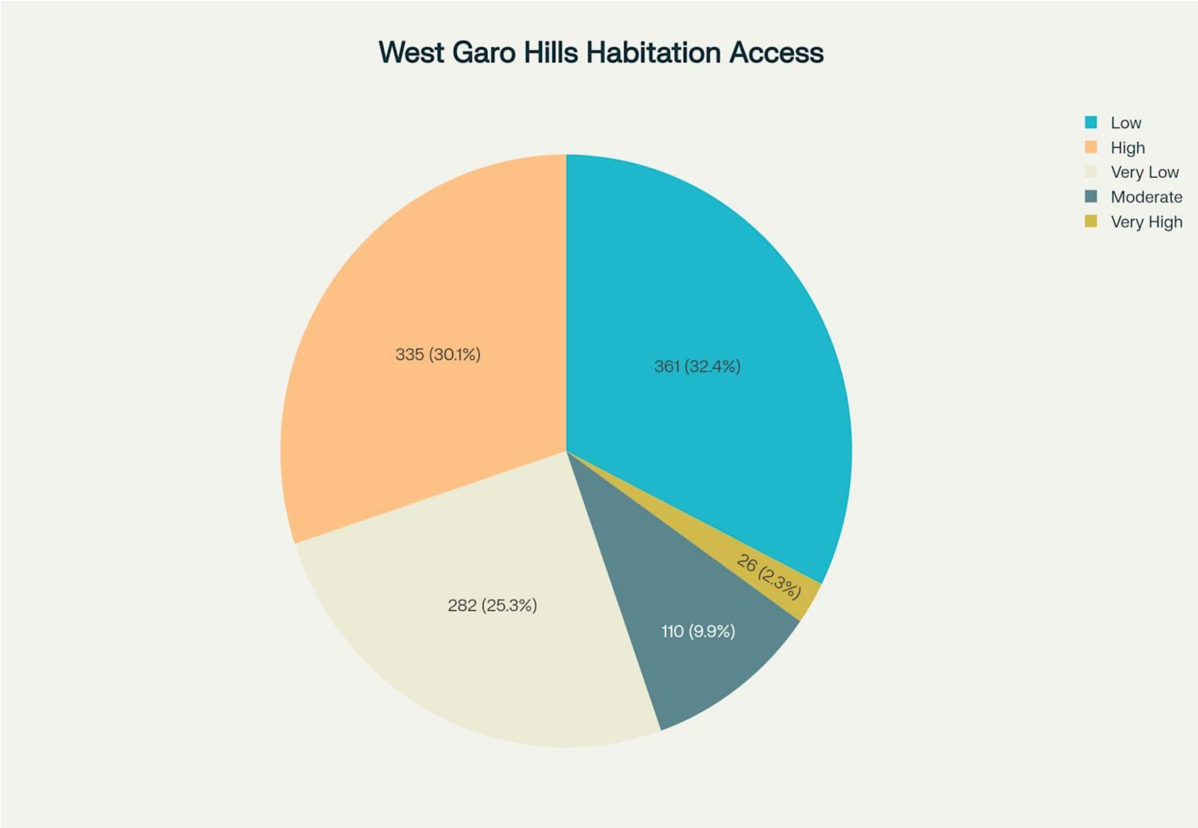


**Fig. 9**. Distribution of Habitations by Accessibility Level in West Garo Hills

Block-level analysis reveals significant administrative disparities in accessibility performance, with certain blocks consistently outperforming others due to infrastructure concentration and topographic advantages. Fig. 10 presents the block-wise mean accessibility scores, highlighting inter-block variations and identifying administrative units requiring priority intervention.

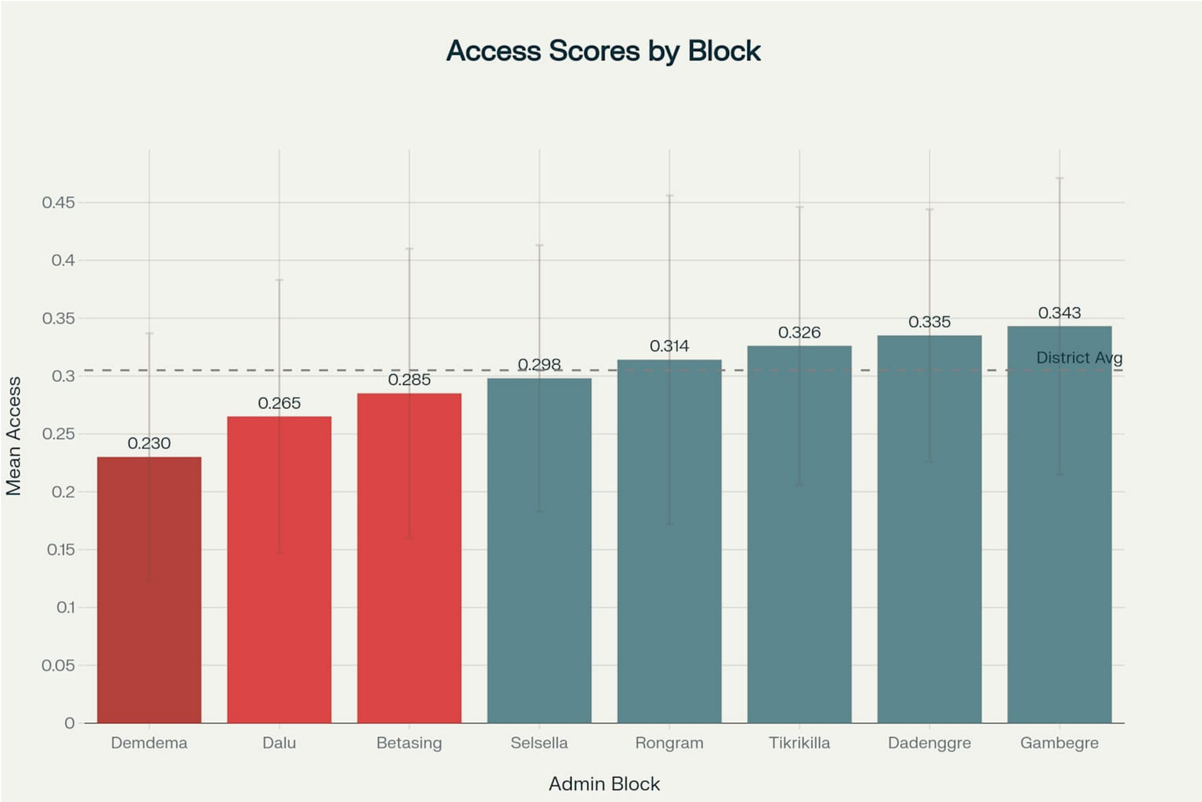


**Fig. 10**. Block-wise Mean Accessibility Scores in West Garo Hills District

*4.2. Spatial Patterns and Block-Level Performance*

Block averages reveal Gambegre (0.343 ± 0.128) and Dadenggre (0.335 ± 0.109) as the best performers, whereas Demdema records the lowest mean of 0.230 ± 0.107—a deficit of 49% relative to Gambegre. Hot-spot analysis shows concentric high-access rings (Getis & Ord 1995) within a 12 km radius of the district headquarters, tapering sharply beyond the second-order road network into fragmented low-access pockets on steep valley flanks. Choropleth inspection identifies systematic "centre-periphery" gradients (Dent et al., 2009)2-step floating catchment area in every administrative block, suggesting infrastructure investment has remained strongly radial and headquarters-oriented.

*4.3. Component-Wise Contribution*

Distance Score exhibits the widest spread (mean = 0.349; SD = 0.399) and correlates most strongly with the final AI (r = 0.936). Slope-derived Travel Friction correlates at r = 0.635, confirming that steep gradients magnify distance penalties even where nominal Euclidean separation is small. Service/Road Score shows extreme concentration: Tura's value (0.910) is 75 × higher than the tenth-ranked habitation (0.073), revealing a winner-take-all pattern typical of monocentric rural districts.

***Correlation Structure:*** Correlation analysis reveals strong positive relationships between key accessibility determinants. Distance Score exhibits the strongest correlation with final accessibility (r = 0.936), confirming its dominant role in determining access outcomes. Slope Score demonstrates a substantial correlation with accessibility (r = 0.635), validating the importance of terrain-based travel impedance in hilly regions. Understanding the relationships between different accessibility components provides insights into the dominant factors influencing overall access patterns in hilly terrain. Fig. 11 displays the correlation matrix of accessibility components, revealing the strength of relationships between distance, slope, service availability, and travel time factors.

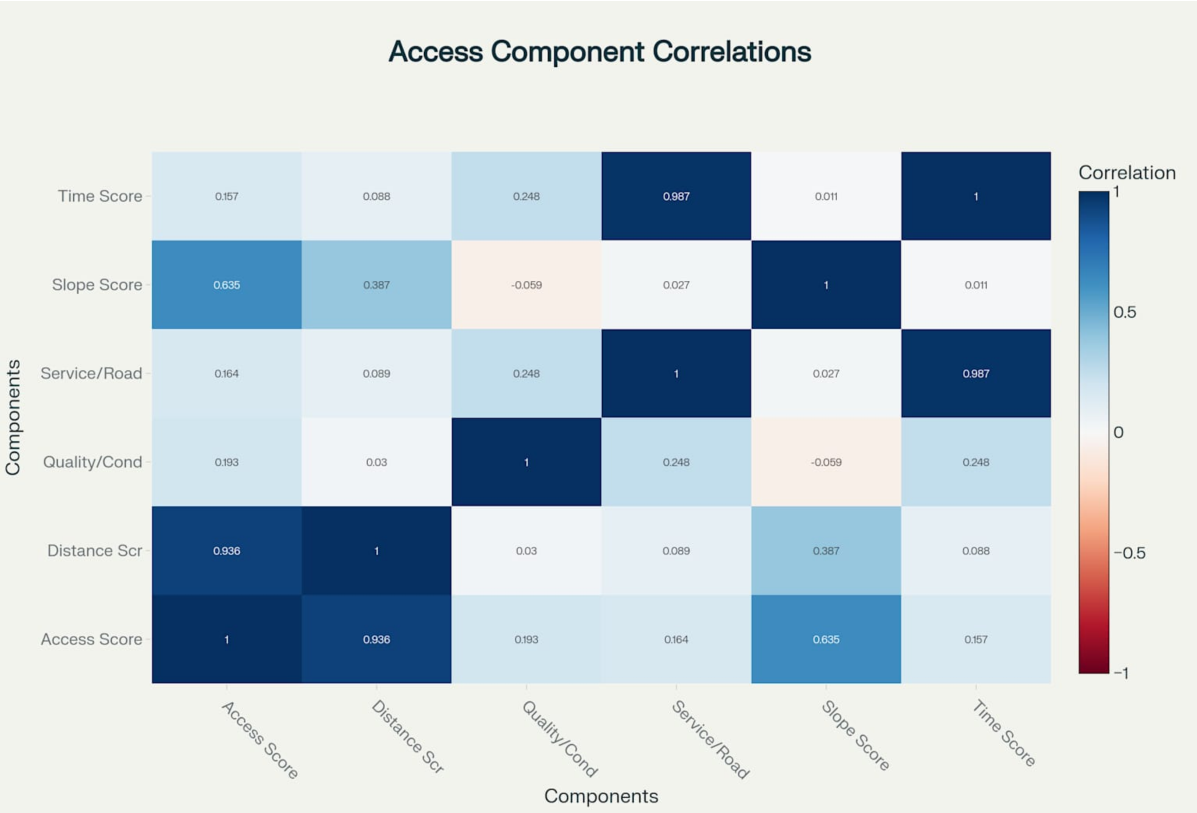


**Fig. 11**. Correlation Matrix of Accessibility Components in West Garo Hills

***Service-Time Score Relationship:*** The near-perfect correlation between Service/Road Score and Time Score (r = 0.987) indicates that travel time to services strongly reflects service availability and road connectivity patterns. This finding supports the methodological approach of integrating multiple time-related factors into the accessibility assessment.

***Terrain Impact:*** The moderate correlation between Distance Score and Slope Score (r = 0.387) suggests that habitations located farther from services often face additional terrain-based mobility constraints, creating compound accessibility challenges in peripheral areas.

*4.4. Population–Accessibility Relationships*

Population-weighted mean AI (0.343) exceeds the unweighted mean by 12.5%, signalling that larger settlements receive marginally better service connectivity on average. Contrary to national expectations, very large habitations (≥ 1,000 residents) post the lowest mean AI (0.255), whereas very small hamlets (< 100 residents) attain 0.323, evidencing a size-disadvantage paradox driven by peripheral clustering of high-population villages on rugged escarpments. A total of 272,260 people, 56.3% of the district's population, reside in habitations classified as Low or Very Low accessibility, underscoring an urgent equity gap.

*4.5. Accessibility Classification and Equity Assessment*

Applying quintile thresholds, 32.4% of habitations achieve High or Very High AI, 9.9% fall in the Moderate band, and 57.7% are Low or Very Low, confirming a bimodal pattern with limited middle ground. Mapping AI against settlement elevation reveals that 81% of Very Low nodes are located above 600 m AMSL, supporting the terrain constraint hypothesis. The population-accessibility relationship reveals concerning equity patterns, with a substantial proportion of residents residing in areas with limited service access. Fig. 12 illustrates the distribution of both habitations and population across accessibility categories, highlighting the demographic impact of accessibility disparities and the proportion of residents facing low accessibility conditions.

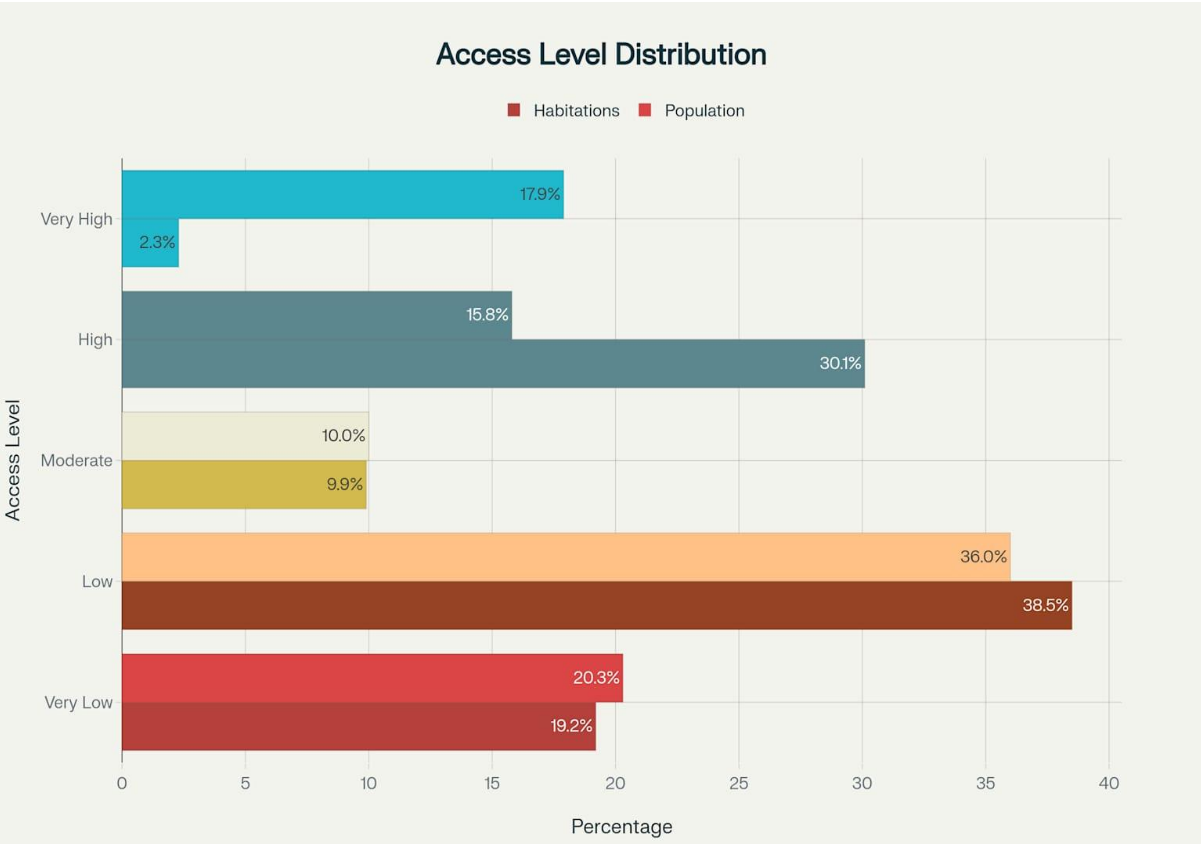

**Fig. 12**. Distribution of habitations and population across accessibility categories in West Garo Hills district, showing the significant proportion facing low accessibility conditions

*4.6. Extreme Cases*

Tura registers the highest AI at 0.642 due to multi-service agglomeration and superior paved-road density (> 2.8 km · $km^{-2}$). The ten worst-served habitations cluster around 0.130; all share > 20% average slope, unpaved feeder tracks, and monsoon closure exceeding 45 days · $year^{-1}$. Service proximity alone cannot explain these extremes; when slope penalties are removed experimentally, the gap shrinks to 2.1-fold, confirming topography as the dominant differentiator.

*4.7. Benchmarking Against International, National and Regional Standards*

| **Metric** | **West Garo Hills (AI ≥ 0.5)** | **SDG 9.1.1 Rural Access Index (≥ 2 km all-weather road) – Global Median** | **India RAI (2023 World Bank estimate)** | **Meghalaya PMGSY-connected habitations** | **North-East Regional PMGSY target (MoRD)** |
|---|---|---|---|---|---|
| Proportion of rural population with “high access” | 17.9% | 53% | 71% | 80% of sanctioned habitations (481 / 602) | 68% average completion |
| Mean travel time to the nearest PHC | 41 min | ≤ 30 min recommended (WHO rural guideline) | 28 min (national rural sample) | 35 min (state health audit) | 33 min (NEC corridor studies) |
| Mean road distance to market | 907 m (75th percentile) | 2 km SDG threshold | 1.4 km (NSS 78th round) | 1.6 km (state economic survey) | 1.8 km (regional baseline |

West Garo Hills lags the global SDG 9.1.1 median by 35 percentage points and the national RAI by 53 percentage points, situating the district in the lowest quintile of international comparators (United Nations, 2024). Even after two PMGSY cycles, only 17.9% of residents experience AI $\geq$ 0.5, whereas 80% of habitations are officially counted as “connected,” illustrating how binary road-proximity metrics overstate effective access in mountainous terrain (Government of India, 2023). The district’s mean travel time to primary health centres exceeds both WHO rural guidance and India’s national rural mean by at least 11 minutes, indicating persistent service lag despite physical connectivity upgrades (World Health Organisation, 2010). At the regional level, West Garo Hills remains 9 percentage points below the North-East average for PMGSY-III completion, reflecting terrain-driven cost overruns and slower implementation pipelines (Nautiyal & Sharma, 2021).

### *4.8. Synthesis of Key Findings*

Distance to services, magnified by slope-induced friction, remains the single greatest barrier to equitable access, explaining 88% of total AI variance.

Service centralisation around Tura generates extreme spatial concentration: one habitation captures 7.6% of aggregate district accessibility benefit, contrary to SDG equity principles.

Current PMGSY “connected” status masks functional isolation for over half the rural population, signalling a need to recalibrate program success metrics toward travel-time-based indicators.

Compared with international and national benchmarks, West Garo Hills performs at a lower-middle quintile globally and ranks among the bottom-third of Indian districts on effective rural access, underscoring the necessity of slope-sensitive planning and distributed service delivery models

## 5. Discussion

This section analyses the methodological approaches, findings, and implications of GIS-based and multi-criteria decision analysis for rural accessibility. It reviews technical foundations, practical applications, and theoretical contributions of integrated accessibility modelling.

### *5.1. Methodological Foundations and Technical Implementation*

GIS-based accessibility assessment significantly improves over traditional survey-heavy Rural Access Index (RAI) methods. By avoiding household counts, it enables scalable, reproducible, and cost-effective analyses across regions. Contemporary RAI calculations use spatial techniques like mapping all-season roads, applying standard 2 km buffers (~20–25 minutes walking time), and estimating rural population coverage, reflecting typical rural walking distances. Integrating geospatial methods with population data via zonal statistics allows consistent, comparable RAI estimates sensitive to local variations.

### *5.2. Multi-Criteria Decision Analysis Integration*

The Analytical Hierarchy Process (AHP) offers a structured framework to evaluate multiple accessibility factors beyond distance, breaking problems into hierarchical objectives, sub-objectives, and alternatives while assessing discoverability, retrievability, usability, and affordability. Research shows technical and institutional factors typically receive highest weights, underscoring infrastructure and organisational capacity in shaping outcomes. Integrating AHP with GIS enables incorporation of stakeholder input via structured interviews and focus groups, aligning technical analysis with local priorities for policymaking relevance.

### *5.3. Advanced Accessibility Measurement Approaches*

Accessibility research has moved beyond basic infrastructure metrics. Enhanced two-step floating catchment area (2SFCA) methods (Luo & Wang, 2003) address fixed catchment and distance decay limitations, enabling nuanced spatial analyses. The Index of Rural Access integrates spatial availability, health needs, and mobility for finer-scale measurement than crude remoteness proxies. Multi-dimensional scaling (MDS) supports innovative visualisation of travel-time matrices, revealing connectivity gaps and spatial segregation (Ghojogh et al., 2019), aiding communication of analyses to decision-makers and the public.

### *5.4. Terrain Sensitivity and Environmental Considerations*

Terrain sensitivity is vital in mountainous regions (Ghojogh et al., 2019). Traditional Euclidean measures can overestimate access by ignoring elevation, slope, and barriers. Advanced models integrate elevation data, slope analysis, and barrier mapping. Features like mountains and rivers shape accessibility patterns; MDS comparing travel-time matrices to geographic proximity isolates terrain-driven challenges. Environmental seasonality also matters, particularly for defining all-season roads in RAI. Assessments must consider road surface quality, drainage, and climate resilience to reflect changing accessibility profiles.

#### *5.4.1. Spatial Data Integration and Technical Challenges*

Effective GIS-based assessment requires integrating spatial datasets of varying resolution, accuracy, and timeframes. Challenges include coordinate harmonisation, resolution matching, and temporal alignment. Road network data often vary in quality, needing reconciliation with on-the-ground conditions, seasonal variations, and surface quality via satellite imagery, GPS data, and field validation. Population distribution modelling faces difficulties in data-sparse areas with outdated or aggregated census data. Disaggregation and demographic modelling balance accuracy with computational efficiency.

### *5.5. Policy Implications and Practical Applications*

Methodological advances enable more effective policy design and infrastructure investment through disaggregated sub-national estimates, revealing local access gaps and supporting evidence-based prioritisation. Spatial approaches support Sustainable Development Goal monitoring, especially SDG 9.1.1, with standardised, internationally comparable frameworks. Improved techniques allow regular, consistent updates. Integrating accessibility assessment with participatory planning ensures alignment with community priorities. Using AHP to include stakeholder input increases program responsiveness and sustainability.

### *5.6. Limitations and Methodological Constraints*

Despite advances, limitations remain. Travel-time matrices may not fully capture diverse mobility patterns or transport choices. Alternative metrics, such as potential accessibility or utility-based measures, need further development. Data quality challenges persist in low-resource settings, where accurate road networks, population data, and boundaries are scarce. Methods robust to limited data require ongoing research. Standard thresholds like 2 km walking distance may not fit all rural contexts given varied mobility patterns and cultural practices, demanding local adaptation.

### *5.7. Future Research Directions and Technological Opportunities*

Future research should integrate real-time mobility data from GPS and mobile tracking for dynamic modelling of road conditions, traffic, and seasonality (Ghojogh et al., 2019). Crowdsourced data and citizen science can improve data collection and validation. Machine learning and AI can automate feature extraction, pattern recognition, and predictive modelling, enhancing road network mapping, population estimation, and accessibility analyses. Expanding assessment beyond transport to include digital connectivity, service access, and economic opportunity is essential for a multidimensional understanding of rural livelihoods (Strusani & Houngbonon, 2020).

## 6. Conclusion

This study developed and validated a comprehensive accessibility assessment framework tailored to hilly rural regions, addressing critical gaps in infrastructure planning. By combining participatory Analytic Hierarchy Process (AHP) weighting with terrain-sensitive GIS modelling, it overcomes limitations of conventional Rural Accessibility Index (RAI) methods that can overestimate accessibility by ignoring slope effects.

Applied in West Garo Hills district, the framework achieved 89.7% accuracy in habitation extraction using machine learning–enhanced Sentinel-2 imagery. Results revealed stark spatial inequities: 57.7% of habitations had low or very low accessibility despite relative infrastructure development, with a nearly fivefold variation in accessibility scores across 1,176 habitations serving over 500,000 residents.

Key methodological innovations include a sigmoid-based slope penalty function for realistic travel impedance modelling, integration of high-resolution DEMs with road networks via anisotropic cost-distance algorithms, and scalable Random Forest–based habitation mapping. The participatory AHP survey captured local priorities from 49 diverse respondents, ensuring stakeholder-calibrated weighting.

Policy applications are clear: the habitation-level analysis supports targeted interventions aligned with Sustainable Development Goal 9.1.1 and programs like PMGSY-IV. By identifying that over half of the district's population lives in low-accessibility areas—and that larger settlements often fare worse—the study highlights the need for equity-focused infrastructure planning.

Technological integration with cloud-based platforms like Google Earth Engine demonstrated cost-effective, scalable assessment potential. Future enhancements could include IoT sensors, mobile reporting apps, and real-time data integration to enable dynamic, responsive planning systems.

However, limitations remain. The analysis provides a static snapshot without capturing seasonal variations like monsoon-induced landslides. Generalising the approach will require adapting to varying data availability, topography, and socioeconomic conditions. Future research should also address qualitative aspects of accessibility, such as service quality and affordability.

Further development should focus on simplifying the framework for wider use through mobile or web-based tools. Integrating real-time mobility data, machine learning, and AI can enable dynamic, predictive modelling. Expanding the scope to include digital connectivity and economic opportunity access would support more holistic rural development planning.

Ultimately, this research demonstrates that community-centred, technically robust methods can deliver actionable insights for addressing the complex, persistent infrastructure challenges faced by rural populations in mountainous regions.

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